Md Zahidul Islam, Ph.D.

Assistant Professor, School of Electrical, Computer & Biomedical Engineering
Southern Illinois University Carbondale, IL, USA

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Professional Appointments/ Employment

Tenure-Track Assistant Professor, ECE — Aug 2025 – Present
Southern Illinois University Carbondale — IL, USA

- Teaching graduate and undergraduate courses in power systems.
- Leading the Power Systems Research Lab at SIUC.
- Supervising Ph.D. students and undergraduate researchers.
- Serving on the Electrical Engineering Program Committee.

Research Assistant — Jan 2021 - Aug 2025
New York University and University of Massachusetts Lowell — NY and MA, USA

Research Intern, Optical Networking & Sensing, NEC LAB — May 2024 - Aug 2024
NEC Laboratories America, Inc. — NJ, USA

Graduate Intern, Power Systems Engineering Center (PSEC), NREL — May 2023 - Aug 2023
National Renewable Energy Laboratory (NREL) — CO, USA

Education

Ph.D. in Electrical and Computer Engineering — Jan 2021 - Aug 2025
New York University (NYU) — New York, NY

- **Dissertation**: Integrating Smart Sensing and Data-Driven Decision Making Toward an Intelligent and Resilient Cyber-Physical Power System (Advisor: Dr. Yuzhang Lin)
- **Relevant Courses**: Power System Operation & Control, Deep Learning, Smart Power Distribution and Distributed Energy Resources, Reinforcement Learning, Communication Network Design, and Network Security

M.S. in Electrical and Electronic Engineering — Oct 2017 - Jan 2021
Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh

- **Thesis**: Adaptive Clarke transformation-based three-phase Phase-Locked-Loop (PLL) under amplitude and phase unbalances in presence of harmonics (Advisor: Dr. Md. Shamim Reza)

B.S. in Electrical and Electronic Engineering — Feb 2013 - Sept 2017
Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh

Teaching Experience

Southern Illinois University Carbondale (SIUC)
Assistant Professor

- **ECE 587: Modern Power System Operation** — Fall 20251
- **ECE 488 / ECE 588: Power System Engineering** — Spring 2026

Daffodil International University (DIU), Dhaka, Bangladesh
Lecturer, Department of Electrical and Computer Engineering — Instructor of Record

- **Computer Programming Language** — Spring 2019
- **Electrical Properties of Materials** — Spring 2019

Student Advising & Supervision

Biswas Rudra Jyoti Arka — Ph.D. student, SIUC; *Topic:* AI-driven automation in power distribution systems; **Role: Supervisor (2025–Present)**

Md. Sayed Hasan Rifat — Ph.D. student, SIUC; *Topic:* Cybersecurity in cyber-physical power systems; **Role: Supervisor (2025–Present)**

Mahin Shahriar — Ph.D. student, New York University; *Topic:* Granular visibility of distribution systems; **Role: Co-Advisor (2025–Present)**

Janith Dassanayake — Ph.D. student, SIUC; *Topic:* Wireless communication research with Dr. Gayan Baduge; **Role: Dissertation Committee Member (2025)**

Isaiah Duckworth and **Samuel Vancil** — Undergraduate students, SIUC; *Project:* Solar power–assisted car design for regional competition; **Role: Undergraduate RSO Advisor (2025–Present)**

Publications

Book Chapters

[B1] Y. Lin, V. M. Vokkarane, **M. Z. Islam**, and S. N. Edib, "Resilient Sensing and Communication Architecture for Microgrid Management," Microgrids: Theory and Practice, Wiley-IEEE Press, 2024.

Journal Papers

[J9] W. Zhang, Y. Lin, **M. Z. Islam**, H. Huang, "Neuro-Physics Hybrid State Estimation of Distribution System with Smart Meter Voltage Measurements", IEEE Transactions on Smart Grid, 2026 (accepted).

[J8] **M. Z. Islam**, Y. Lin, V. M. Vokkarane, "Disaster-Resilient Cyber-Physical Distribution System Reconfiguration and Dynamic Networked Microgrid Formation under Intermittent Generation", IEEE Transactions on Industry Applications, vol. 62, no. 2, pp. 3459-3471, March-April 2026.

[J7] **M. Z. Islam**, Y. Lin, W. Zhang, "Smart Meter Scheduling for Data-Driven Granular Customer Voltage Visibility", IEEE Transactions on Smart Grid, vol. 17, no. 1, pp. 832-844, Jan. 2026.

[J6] **M. Z. Islam**, Y. Lin, V. M. Vokkarane, "Observability-aware Resilient PMU Networking", IEEE Transactions on Power Systems, vol. 40, no. 1, pp. 218-230, Jan. 2025.

[J5] **M. Z. Islam**, Y. Lin, V. M. Vokkarane, N. Yu, "Robust Learning-based Real-time Load Estimation Using Sparsely Deployed Smart Meters with High Reporting Rates", Applied Energy, Volume 352, 2023, 121964.

[J4] **M. Z. Islam**, S. N. Edib, V. M. Vokkarane, Y. Lin and X. Fan, "A Scalable PDC Placement Technique for Fast and Resilient Monitoring of Large Power Grids," in IEEE Transactions on Control of Network Systems, vol. 10, no. 4, pp. 1770-1782, Dec. 2023.

[J3] **M. Z. Islam**, Y. Lin, V. M. Vokkarane, and V. Venkataramanan, "Cyber-physical Cascading Failure and Resilience of Power Grid: A Comprehensive Review", Frontiers in Energy Research, 11, p.1095303.

[J2] **M. Z. Islam**, M. S. Reza, M. M. Hossain and M. Ciobotaru,"Accurate Estimation of Phase Angle for Three-Phase Systems in Presence of Unbalances and Distortions," in IEEE Transactions on Instrumentation and Measurement, vol. 71, pp. 1-12, 2022, Art no. 9001712.

[J1] **M. Z. Islam**, M. S. Reza, M. M. Hossain and M. Ciobotaru, "Three-Phase PLL Based on Adaptive Clarke Transform Under Unbalanced Condition," in IEEE Journal of Emerging and Selected Topics in Industrial Electronics, vol. 3, no. 2, pp. 382-387, April 2022.

Conference Papers

[C10] M. Shahriar, **M. Z. Islam**, W. Zhang, and Y. Lin, "Making Low-Voltage Networks Granularly Visible: Strategic Metering and Data-Driven Estimation via Graph Analytics and Learning," ACM e-Energy, 2026 (accepted).

[C9] B. R. J. Arka, **M. Z. Islam***, Y. Lin, V. M. Vokkarane, and J. Zhao, "Communication Network-Aware Missing Data Recovery for Enhanced Distribution Grid Visibility", IEEE Power & Energy Society General Meeting (PESGM), 2026 (accepted).

[C8] **M. Z. Islam***, Y. Lin and V. M. Vokkarane,"Cyber Security Constrained Economic Dispatch for Resilient Power System Operation," 2025 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), North York, ON, Canada, 2025, pp. 1-6.

[C7] Y. Ding, **M. Z. Islam**, J. Shiau, A. D. Amico, Y. Tian, Z. Jiang, S. Ozharar, T. Wang, and Y. Lin, "Resilient DFOS Placement Strategy for Power Grid Monitoring: Integrating Fiber and Power Network Dependencies," 29th International Conference On Optical Fibre Sensors (OFS), Porto, Portugal, 2025, Vol. 13639, pp. 1169-1172.

[C6] **M. Z. Islam***, Y. Ding, Y. Tian, T. Wang, Y. Lin, "Integration of Fiber Optic Sensing and Sparse Grid Sensors for Accurate Fault Localization in Distribution Systems", 2025 IEEE Power & Energy Society General Meeting (PESGM), Austin, TX, USA, 2025, pp. 1-5.

[C5] P. Christou, **M. Z. Islam***, Y. Lin, and J. Xiong, "LLM4DistReconfig: A fine-tuned large language model for power distribution network reconfiguration," 2025 Annual Conference of the Nations of the Americas Chapter of the ACL (NAACL), Albuquerque, New Mexico, USA, pp. 4136-4155.

*Corresponding author.

- **[C4]** **M. Z. Islam**, W. Zhang, Y. Lin, "Learning-based Customer Voltage Visibility with Sparse High-Reporting-Rate Smart Meters," 2024 IEEE Power & Energy Society General Meeting (PESGM), Seattle, WA, USA, 2024, pp. 1-5.
- **[C3]** **M. Z. Islam**, Y. Lin, V. M. Vokkarane, Y. Yao and F. Ding, "Cyber-Physical Reconfiguration for Disaster Resilience of Power Distribution Systems," 2023 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), Glasgow, United Kingdom, 2023, pp. 1-6.
- **[C2]** **M. Z. Islam**, V. M. Vokkarane and Y. Lin, "PMU Network Routing for Resilient Observability of Power Grids," ICC 2023 - IEEE International Conference on Communications, Rome, Italy, 2023, pp. 4584-4590.
- **[C1]** E. Meriaux, D. Koehler, **M. Z. Islam**, V. Vokkarane and Y. Lin, "Performance Comparison of Machine Learning Methods in DDoS Attack Detection in Smart Grids," 2022 IEEE MIT Undergraduate Research Technology Conference (URTC), Cambridge, MA, USA, 2022, pp. 1-5.

Manuscripts Under Review

- **[U1]** **M. Z. Islam**, Y. Yao, F. Ding, Y. Lin, "Risk-Aware Measurement Synchronization and Recovery for DSSE with Heterogeneous Data Sources", IEEE Transactions on Instrumentation & Measurement (under review).

Selected and Ongoing Research Projects

Granular Visibility-Aware Control Strategies for Distribution Grid-Edge Resources (2025–Present)

- Developing distributed control strategies for grid-edge resources that leverage prior work on high-resolution network visibility.

Application of Large Language Models (LLMs) in Power Systems (2025–Present)

- Developing LLM-based decision-support tools for power system planning and operation.
- Integrating domain knowledge and optimization frameworks with LLM reasoning for explainable decision-making.
- Assessing security and reliability challenges of LLM-assisted grid operations.

DOE CyberCARED: Northeast University Cybersecurity Center for Advanced and Resilient Energy Delivery (2025–Present)

- Developing online attack detection and localization strategies using cyber-physical parameter fusion.

Cybersecurity in Power Systems (2025–Present)

- Developing a moving target defense strategy against false data injection attacks in power systems.

Services & Professional Activities

- **Electrical Engineering Committee**: Serving as a member of Electrical Engineering Committee at SIUC to modify its curricular and suggest new inclusions to satisfy ABET accreditation.
- **Journal & Conference Reviewer**: Reviewed for IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, IEEE transactions on Sustainable Energy, Reliability Engineering & Systems Safety, IEEE GLOBECOM, IEEE SmartGridComm, IEEE PES General Meeting, IEEE PES ISGT.
- **Outreach**: Presented and demonstrated research activities of the SIUC Power Systems Lab to middle and high school students and their families as part of SIUC outreach programs.
- **Senator, Graduate Bangladeshi Student Association at UML (2022-2023)**: Represented the student body, communicated with the administration, participated in decision-making, and organized student programs.
- **Event Coordinator (2016-2017) and Advisor (2017-2021), Kazipur Math Festival**: Organized annual math festivals for underprivileged students in remote areas of Bangladesh and provided mentorship for national math Olympiads.

Talks & Presentations

- **State Estimation**: "State Estimation in Power Distribution Systems under Heterogeneous Data Sources." Invited talk, National Renewable Energy Laboratory, Aug. 2023.
- **Fault Localization**: "Integrating Electric Point Sensors and Fiber Sensing for Fault Localization in Power Networks." Invited talk, NEC Laboratories America, Aug. 2024.
- **Voltage Estimation**: "Learning-based Customer Voltage Visibility with Sparse High-Reporting-Rate Smart Meters." IEEE PES General Meeting, Nov. 2024.
- **Resilience & Restoration**: "Disaster Resilience and Service Restoration in Distribution Systems." Invited talk, New York University, Nov. 2024.

- **Forecasting in Power Systems**: "Learning-based Voltage and Demand Forecasting in Power Systems." Invited talk, New York University, Jan. 2024.
- **Sensor Networking**: "Resilient Sensor Networking in Power Systems." Invited talk, University of Massachusetts Lowell, Feb. 2023.

Honors & Awards

- **Travel Grant**, IEEE Power & Energy Society General Meeting 2024
- **Research Excellence Award** 2023
  Awarded for outstanding Master's research, Bangladesh University of Engineering and Technology (BUET).
- **GIS and SCADA Certification** 2020
  Awarded by the Bangladesh Power Management Institute for outstanding performance among 50 nationwide participants.
- **BUET Duke of Edinburgh Award** 2016
- **Bangladesh National Math Olympiad** 2010
  Runner-up.

Industry Engagement

U.S. Utility – Ameren, Illinois: Collaborating with their distribution network in the Carbondale, Illinois area to enhance system resiliency.

US NAVFAC EXWC Microgrid Academy Training: Completed a comprehensive day-long training program focused on microgrid modeling using XENDEE software, exploring real-world Navy microgrid case studies. Gained hands-on expertise in designing resilient microgrid systems optimized for naval operations.

Con Edison Utility in New York City: Engaged in an on-site visit to Con Edison's distribution network, gaining firsthand exposure to the layout and functionality of primary and secondary feeders. Observed operational challenges within the underground distribution system and explored opportunities for collaboration.

Prior Industry Experience

Assistant Engineer, Network Operation and Customer Service. Nov 2019 - Jan 2021
Dhaka Power Distribution Company (DPDC) Dhaka, Bangladesh

- Managed multiple distribution feeders, focusing on improving reliability metrics such as SAIDI and SAIFI.
- Supervised control room operators to ensure efficient network operations.

IT & Cloud Engineer, Data Center IT Infrastructure Design Dec 2017 - Aug 2018
ZTE Corporation Bangladesh Ltd. Dhaka, Bangladesh

- Contributed to the first IV-tier national data center in Bangladesh.
- Validated the Deep Packet Inspection (DPI) system to strengthen network security and optimized the iCache service to enhance intranet speed.

Technical Skills

Programming Languages: MATLAB, Python, C, C++.
Machine Learning, Deep Learning Frameworks: Scikit-learn, TensorFlow, PyTorch, Keras, Ray.
Simulation Tools: OpenDSS, Simulink, RT-LAB(OPAL-RT), PSS/E, PSpace, CYME PSAF, NS-3.